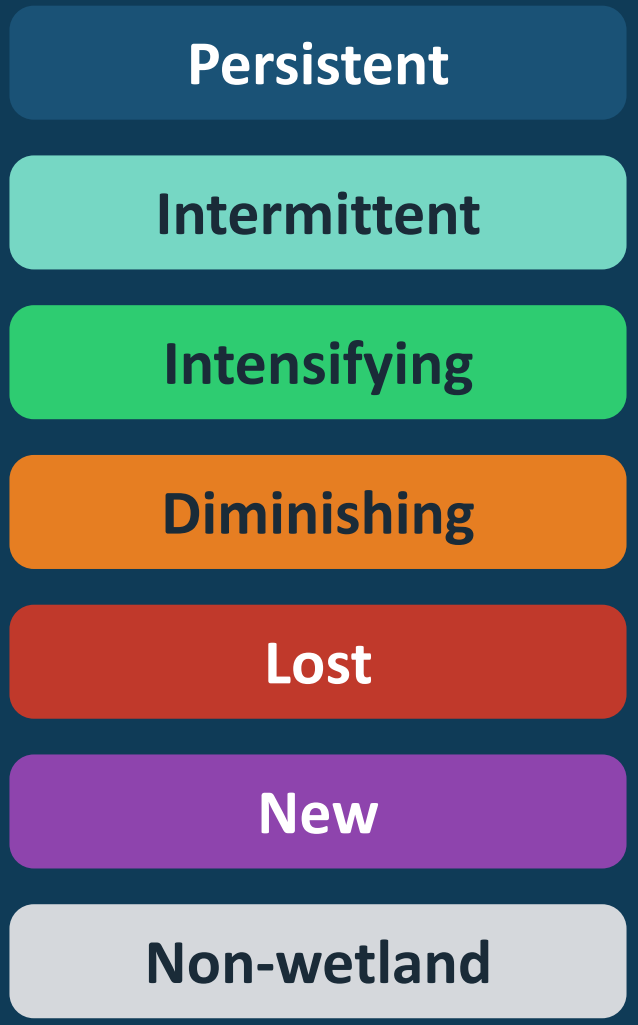
WetlandMapper

From an outline to wetland maps, statistics and GeoTIFFs

Wetlands Indaba 2026 | 90-minute workshop

Conveners: Manudeo Singh and Stephen Tooth (Aberystwyth University), and Ruth-Mary Fisher (SANParks)

`github.com/manudeo/Wetlands_Indaba_2026_workshop`



Why this workshop



What the work needs

Wetland extent and condition tracked over time, for management, restoration priorities and reporting.



What gets in the way

Remote sensing workflows that expect programming skills and software many practitioners cannot easily access.



What you get today

A free notebook. Upload an outline of your wetland, press run, and get maps, numbers and GeoTIFFs.

Where WetlandMapper fits

1

Map and classify

Where are the wetlands, and what kind are they?

Wetness contrast: a pixel that is wet often enough is wetland. Mapped inventories and field mapping give the reference.

Part A

2

Assess the status through time

How healthy is each wetland? Is it shrinking, fragmenting, drying out or being encroached?

Current and historic time scales: wet and dry years per pixel, and what the surface is made of.

Part B

Part C

3

Find the causes

What controls the hydrology: rainfall, connectivity to other waterbodies, the catchment?

Rainfall against wet area is in Part B. Connectivity and catchment analysis are beyond today.

Part B (rain)

Beyond today

What you take home



Your outline

KMZ or KML from Google Earth, GeoPackage, GeoJSON or a shapefile



A satellite record

36 years of Landsat; the last two weeks of Sentinel-2



WetlandMapper

Wet and dry years per pixel; what the wetland is made of



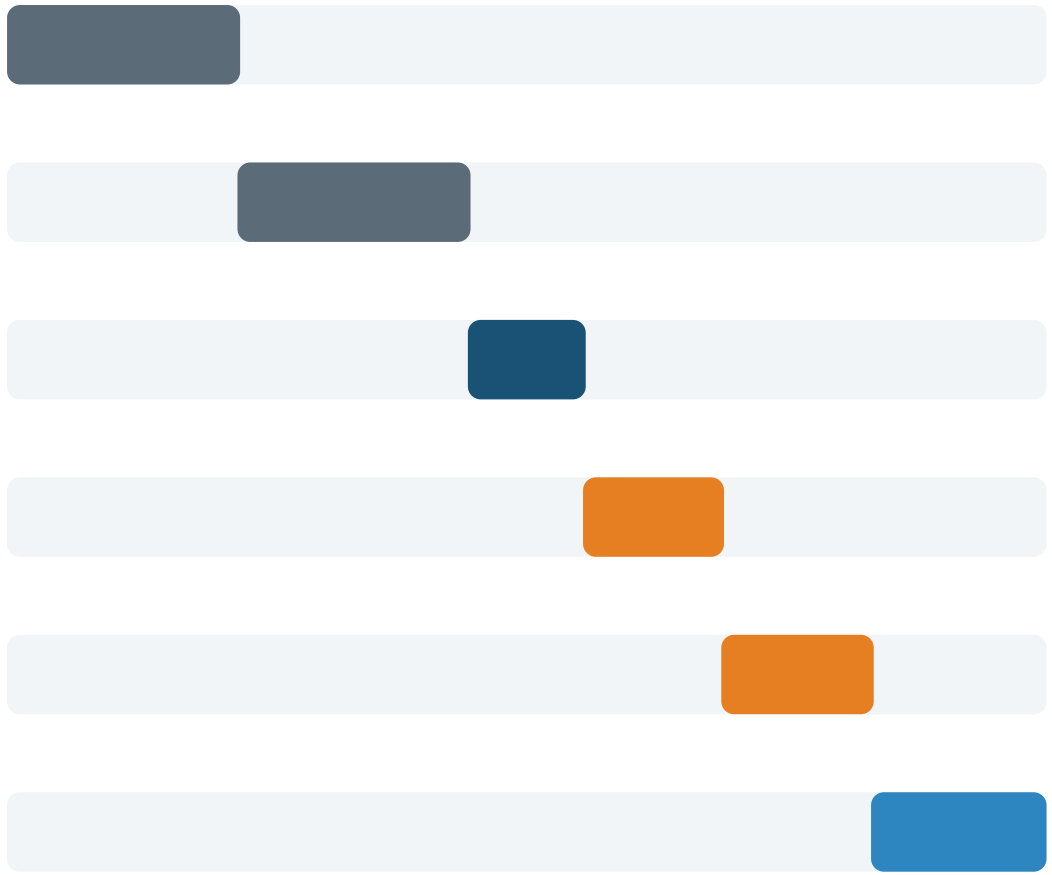
Results

Statistics, figures, GeoTIFFs, wetland polygons

In the zip you download

GeoTIFFs with the class colours embedded | wetland polygons as GeoPackage and KMZ | tables as CSV | figures as PNG | a short report in plain sentences | a run manifest with the settings and software version

Ninety minutes



0:00 — **The science behind the methods**
The slides: what the classes are and where they come from (20 min)

0:20 — **WetlandMapper on GitHub**
The repository, example notebooks, reporting an issue (20 min)

0:40 — **Set up Colab**
Everyone opens the notebook and runs the setup cell (10 min)

0:50 — **The preset notebook, without Earth Engine**
Parts A, B and C on the sample data (10 to 15 min)

~1:02 — **The same areas, with Earth Engine**
Choose your own months; a live map of this week (10 to 15 min)

~1:15 — **Your own region**
Upload an outline and run it, then questions (the rest of the time)

Two questions, two products

How has it behaved?

Wetland dynamics

- Persistent
- Intermittent
- Intensifying
- Diminishing
- Lost
- New
- Non-wetland

Each pixel is classed from the share of years it was wet and from how its first and last years compare. Anything that is not class 0 is wetland.

What is it made of?

Wetland cover types

- Open clear water
- Turbid water
- Submerged vegetation
- Emergent vegetation
- Moist soil

Water, vegetation and turbidity indices are each split into four levels. The three levels form a code (such as W1V1T2) and a lookup gives the class. No training data.

How a pixel gets its dynamics class

W% share of years the pixel was wet

historic wet years in the first N years

recent wet years in the last N years

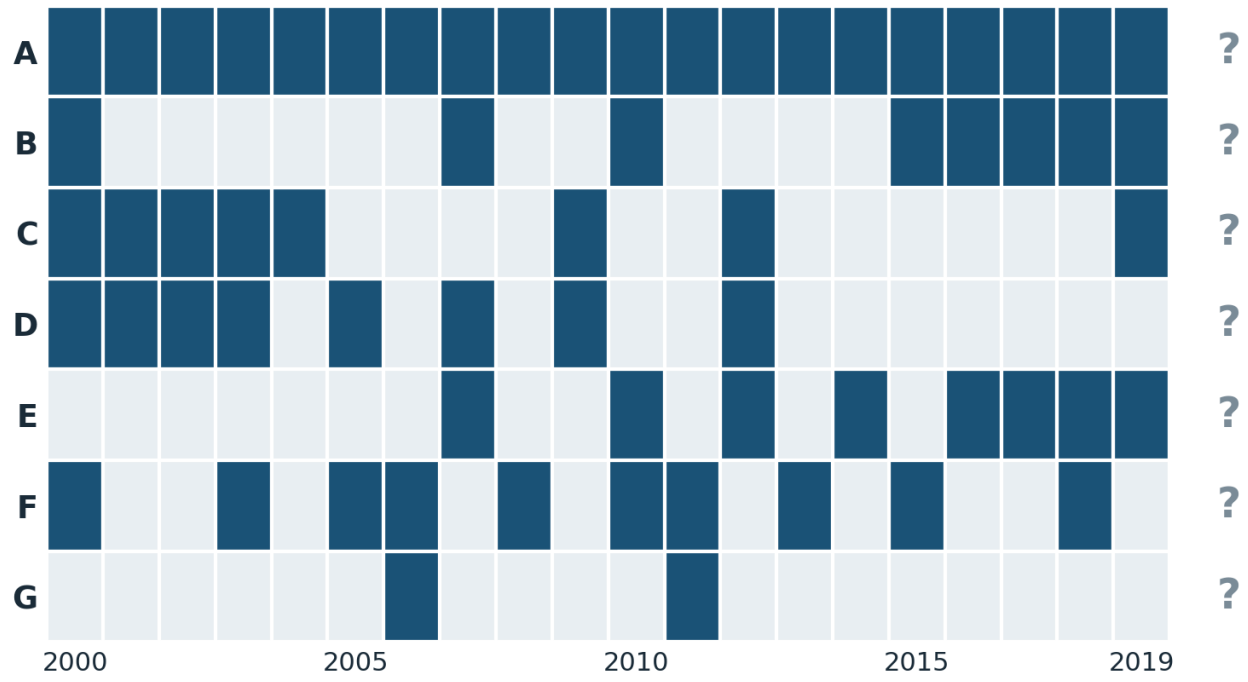
A year is wet when the water index is above 0.

Workshop defaults: 3-year windows, thresholds 25% and 75% (sliders in the notebook). The published method (Singh and Sinha 2022) used 5-year windows and 50% and 90%, and called a pixel New or Lost only if it was wet or dry in all five years. Arid places may need a lower wetness threshold.

Rules are checked in this order; the first match wins

1	**Persistent**	wet in at least 75% of years
2	**New**	dry at the start, wet at the end
3	**Lost**	wet at the start, dry at the end
4	**Intensifying**	wet 25% or more, more wet years recently
5	**Diminishing**	wet 25% or more, fewer wet years recently
6	**Intermittent**	wet 25% or more, no direction
7	**Non-wetland**	wet in less than 25% of years

Predict the class of each pixel



Seven pixels, 20 years

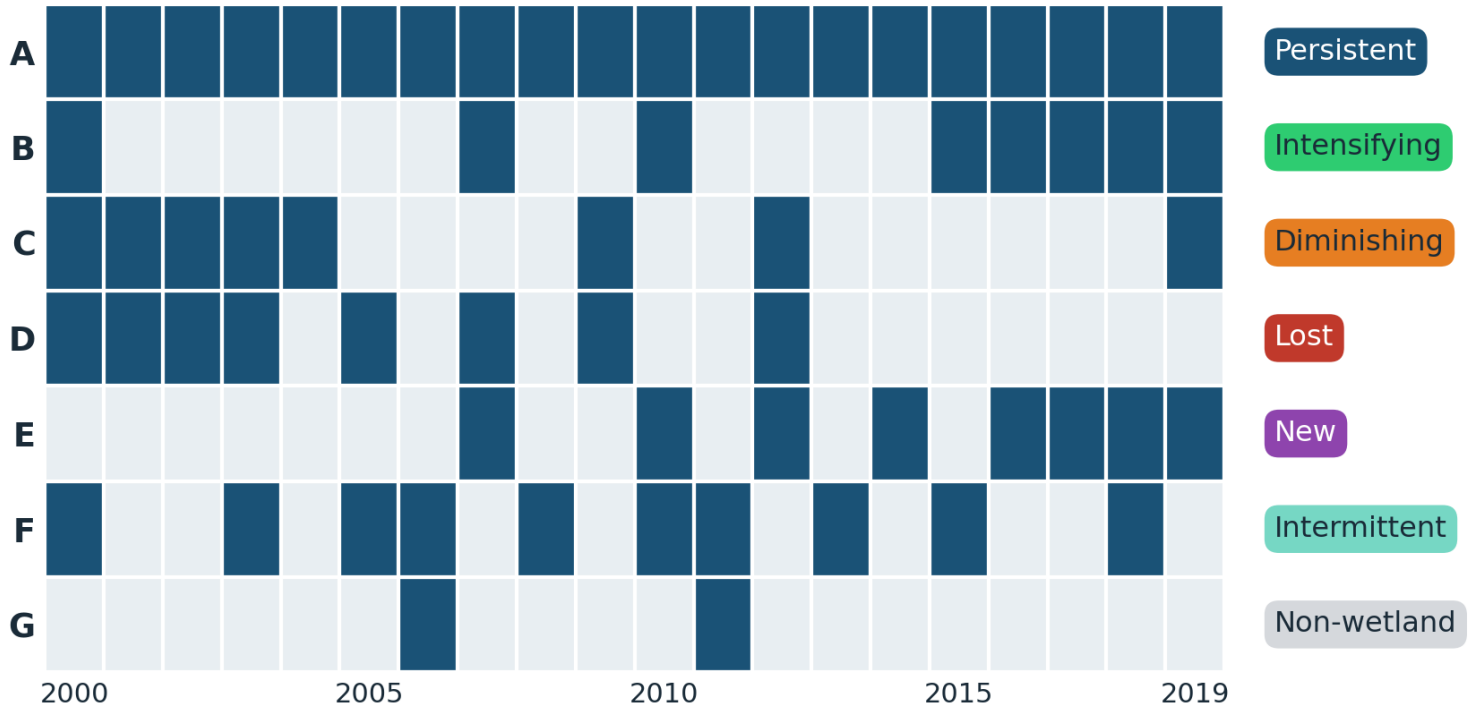
Dark cells are wet years.

Use the rules and the first and last 3 years.

Write your guess for A to G.

```
nYear=3  thresholdWet=25  thresholdPersis=75
```

What the function says

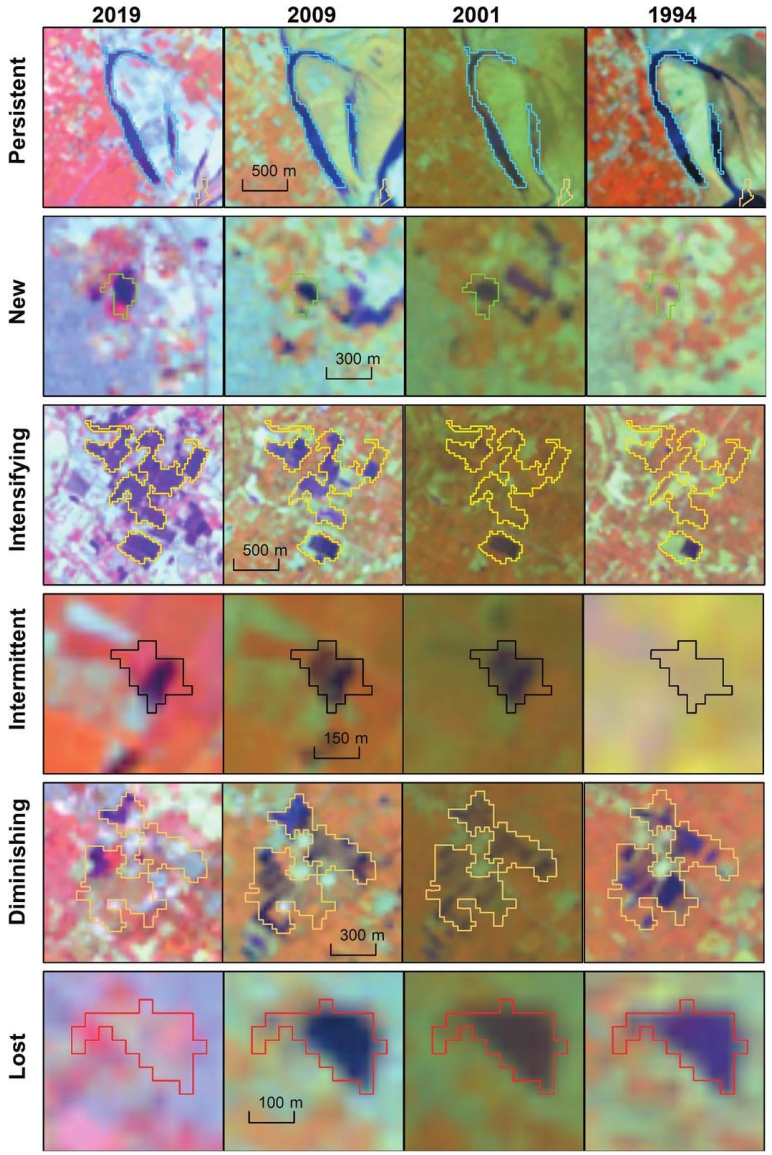


D is Lost, not Diminishing: wet early, dry in the last 3 years, although it was wet 40% of the time.

G is Non-wetland: wet in 2 of 20 years (10%), below the 25% threshold.

B and C have the same W% (40%); the direction of change separates them.

What the six classes look like



Persistent — Wet in nearly every year: the stable core that holds water in dry spells.

New — Not wet at the start of the record, wet in recent years.

Intensifying — Wet more often in recent years than at the start.

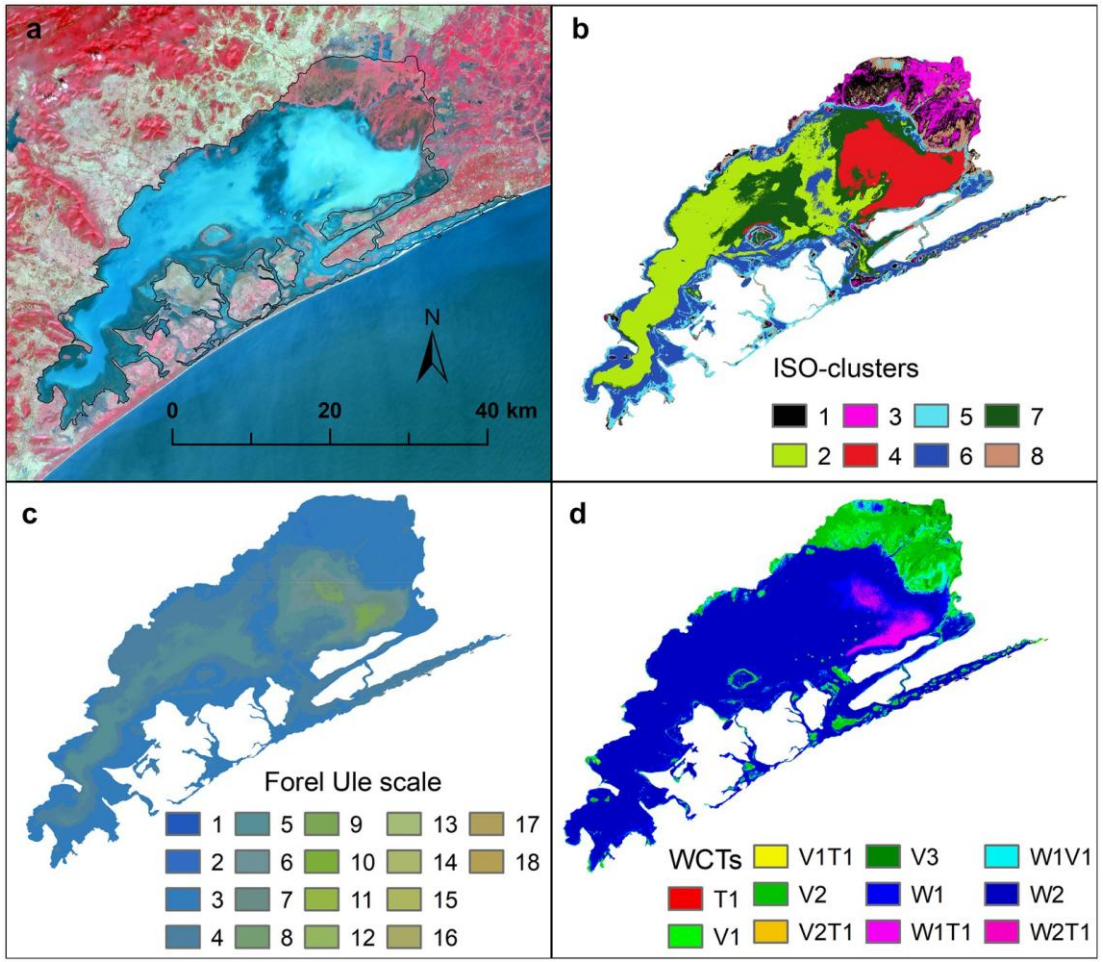
Intermittent — Wet in some years with no direction: episodic wetness.

Diminishing — Wet more often at the start than recently: drying.

Lost — Wet at the start, dry in recent years.

Each row is one wetland; the columns are 2019, 2009, 2001 and 1994 (Landsat, false colour). The outlines mark the mapped wetland. Singh and Sinha (2022), Remote Sensing Letters, Fig. 2.

Why cover types, not clusters or colour scales



Chilika Lagoon, Sentinel-2, 1 February 2021. a, false-colour composite; b, ISO clusters; c, Forel-Ule colour scale; d, cover types. Singh et al. (2022), Environmental Monitoring and Assessment, Fig. 11.

ISO clusters

Distinct patches, but random classes. What a class means cannot be known without extensive field measurements.

Forel-Ule colour scale

Works only for open water. The vegetated north of the lagoon comes out as open water.

Cover types

Distinct classes that include the non-water types, and each class says something about the water, vegetation and turbidity underneath.

If you use QGIS or ArcGIS: the same steps, run for you

In your GIS	In the notebook
Clip to your boundary	The outline you upload sets the area and the statistics
Reproject, set the cell size	GeoTIFFs are written in the local UTM zone; areas are in hectares
Raster calculator: a water index	AWEIsh, MNDWI, NDVI, NDTI from Landsat and Sentinel-2, per year
Reclassify with thresholds	Dynamics classes from the share of wet years; two sliders
Zonal statistics by polygon	One row per polygon in a CSV: area, class areas, wet-area trend
Raster to polygon	Wetland objects as GeoPackage and KMZ, with area and wet frequency
Symbology	The class colours are embedded in every GeoTIFF

The difference is the satellite record: 36 years are fetched and processed for you, and you get the same report for any outline.

Part A: find the wetlands in an area

The Agulhas Plain, 1990 to 2025. Which pixels behave like wetland, and where does the national wetland map agree?

1 Run it

One cell, two sliders: how many wet years count, and the smallest object kept.

2 Read the map

Persistent and intermittent are the stable part. New and lost changed between the first and last years.

3 Read the table

Share of the mapped layer flagged as wetland, by type. A low share is a finding about the data, not necessarily an error.

4 Look at the candidates

Wetland objects that do not touch the mapped layer: places to check in imagery or in the field.

Change the sliders and run again: report the thresholds with any map.

Part B: one wetland through 36 years

A mapped saline pan on the Agulhas Plain (OBJECTID 3462). The same report you get for any outline.

Wet area by year

with rainfall on the same chart

Dynamics map

which parts are always wet, which only in wet years

Cover types by year

open water, vegetation, moist soil through time

Tables and sentences

per-polygon statistics; a trend as a direction, with its caveat

How to read the trend. Thirty-six noisy yearly values can show a direction, not a precise rate. A wetland driven by rainfall can show a trend only because the record starts in a wet or dry run. The notebook says 'no clear trend' when that is what the data show.

Part C: cover types this week, Langebaan

A recent Sentinel-2 map of the Ramsar site, to compare with what you see on your field visit.

Scenes

The last days of Sentinel-2 scenes are listed with their date and time in South African time.

Composite

Cloudy scenes are dropped; the rest give a median composite and a cover-type map.

Your field points

Add a CSV, KMZ or GeoPackage of points to get the class and the three index values at each point.

Context

The areas are compared with the usual wet season, 2016 to 2025, as context only. Landsat and Sentinel-2 differ, and one sensor per time series is advised.

The classes are spectral, not a verified habitat.

Your own area

1 Draw or find an outline

A larger area to search (Part A) or one wetland (Parts B and C), as polygons. In Google Earth: Add, Polygon, Save place as KMZ.

2 Upload it

KMZ, KML, GeoPackage, GeoJSON, zipped shapefile, or a path in Drive. Points and lines are not enough.

3 Check the rainy season

The notebook looks up rainfall at your site and suggests the months for cover types. Gauteng and KZN are not the Western Cape.

4 Run, then download

Statistics, figures, GeoTIFFs and polygons as a zip, or copied to your Drive.

Earth Engine is free for non-commercial use; check its terms for commercial work.

You need

- A Google account
- Earth Engine, with a Google Cloud project ID and the Earth Engine API enabled
- An outline file

No outline? There is an example KMZ to test the route.

No Earth Engine? Parts A to C still run on cached sample data. For your own area, a route without Earth Engine is planned in the package itself.

Reading results: what it sees, what it misses

Sees well

- Open water, and how often it is there
- Change between wet and dry years
- Large wetlands, pans and lagoons

Sees poorly

- Small, vegetated or briefly wet features: springs, seeps, reed beds
- Steep, shaded ground (shadow can look like water)
- Dry climates, where water is too brief for a yearly composite
- Landsat 7 after 2003: stripes leave about a quarter of a scene empty. The published method left it out; the notebook keeps it so that years are not empty

Always report with the map: the index, sensor, composite rule, thresholds, years and software version. The run manifest in your zip has them. Satellite classes are a repeatable screening layer; they do not replace field knowledge.

Beyond today: built in, but not run today



Terrain-adaptive

A DEM masks steep ground, where shadow can look like water. Slope is checked against the wetlands you know, because a slope mask also removes hillslope seeps.

```
dem_mask=True
```



Climate-adaptive

Monthly rainfall chooses which month to sample, because dryland water comes in episodes. It works at monthly resolution and cannot place an event within a month.

```
climate_adaptive=True
```

Both are described in the WetlandMapper documentation and demo notebooks (wetlandmapper.readthedocs.io), with the settings and the checks to make before using them.

Take it with you

Workshop notebook, data and guide

`github.com/manudeo/Wetlands_Indaba_2026_workshop`

WetlandMapper source

`github.com/manudeo/wetlandmapper`

Documentation

`wetlandmapper.readthedocs.io`

Cite the methods

`Singh and Sinha (2022); Singh et al. (2022); Singh (2026)`

Wetland health, an application

`Singh and Sinha (2021), Ecological Indicators 127, 107739`

Conveners: Manudeo Singh and Stephen Tooth (Aberystwyth University), Ruth-Mary Fisher (SANParks)

manudeo.singh@aber.ac.uk

Thanks

SANParks for the Agulhas Plain Ramsar wetlands layer, a subset of the National Wetland Map 5.

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